TEMASEK JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 1

H1 GEOGRAPHY

8834/01

Paper 1

23 August 2024

Additional Materials: Insert

3 hours

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer both questions in **Section B**.
Answer **two** questions in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **4** printed pages.

Section A

Cluster 2 Fieldwork

1 A group of 20 students from Temasek Junior College carried out fieldwork to investigate the attitudes and responses of local residents towards climate change. They focused on the environmental lifestyle choices of the local community in Tampines, a neighbourhood in Singapore. The research question for their investigation was:

'To what extent is the local community in Tampines making lifestyle changes to reduce their carbon footprint?'

In 10 groups of 2, the students approached 100 people in Tampines Regional Centre who seemed like a resident in the area and administered a questionnaire to survey their attitudes towards climate change. The survey was conducted on a Wednesday afternoon after school hours.

Following the questionnaire survey, the students conducted a waste visual assessment and interview by recruiting 10 respondents to gain richer qualitative insights into their waste management practices and requested them to take a photograph of their waste bin (as shown in Resource 4).

Resource 1 shows the map of Tampines, Singapore. Resource 2 shows photographs of Group 1's and Group 2's survey sites in the Tampines neighbourhood. Resource 3 shows results of the questionnaire survey. Resource 4 shows the sorting of waste in one respondent's home.

(a) Explain why the research question is appropriate for the investigation. [3]

(b) With reference to Resources 1 and 2, suggest how the students could minimise potential risks when visiting the Tampines neighbourhood for their investigation. [4]

(c) Explain **three** criteria the students could use to select participants for the waste visual assessment and interview. [3]

(d) Use a suitable data representation technique to represent the tallied data shown in Question 1 of Resource 3. [5]

(e) Using Resources 3 and 4, answer the research question. [5]

(f) Evaluate the students' investigation of residents' attitudes and responses towards climate change. [10]

Section B

Cluster 1 Sustainable Future and Climate Change

Answer both questions

2 Resource 5 shows the city liveability index of cities in 2023. Resources 6A and 6B shows the changes before and after urban reimaging in Birmingham's city centre. Resource 7 shows the resource flows depicting urban metabolism of Birmingham, United Kingdom.

(a) Using Resource 5, describe the distribution of cities based on their city liveability index in 2023. [5]

(b) Using Resources 6A and 6B, contrast the changes shown in the centre of Birmingham. [5]

(c) Using an example, explain how there is both complementarity and tension between sustainable urban development and liveability. [3]

(d) Using Resource 7, describe the main patterns of resource flows depicting urban metabolism in Birmingham. [5]

(e) Explain how the demand placed on the environment by urban areas could result in environmental problems in surrounding areas. [4]

3 Resource 8 shows the average solar radiation and average global surface temperature change, 1910-2010. Resource 9 shows the top emitter countries and their major sources of CO_2 emissions. Resource 10 shows some possible consequences of contemporary climate change.

(a) Using Resource 8, compare the trends in average solar radiation and average global surface temperature change. [4]

(b) Using Resource 9, describe the cumulative contributions of CO_2 emissions by the top emitter countries between 1850-2021 [5]

(c) Explain how the global carbon cycle can be negatively impacted by human activities such as those shown in Resource 9. [4]

(d) Using Resource 10, describe the main patterns of the possible consequences of contemporary climate change. [5]

(e) Explain how climate change could affect **two** groups of people differently in countries at lower levels of development. [4]

Section C

Answer two questions

4 Evaluate the varying success of strategies to manage non-hazardous solid waste across places. [13]

5 Evaluate the success of strategies to address the issues faced by the elderly living in cities. [13]

6 'The uncertainties over future impacts of climate change are the most significant challenge in implementing strategies to respond to contemporary climate change.'

Evaluate this statement. [13]

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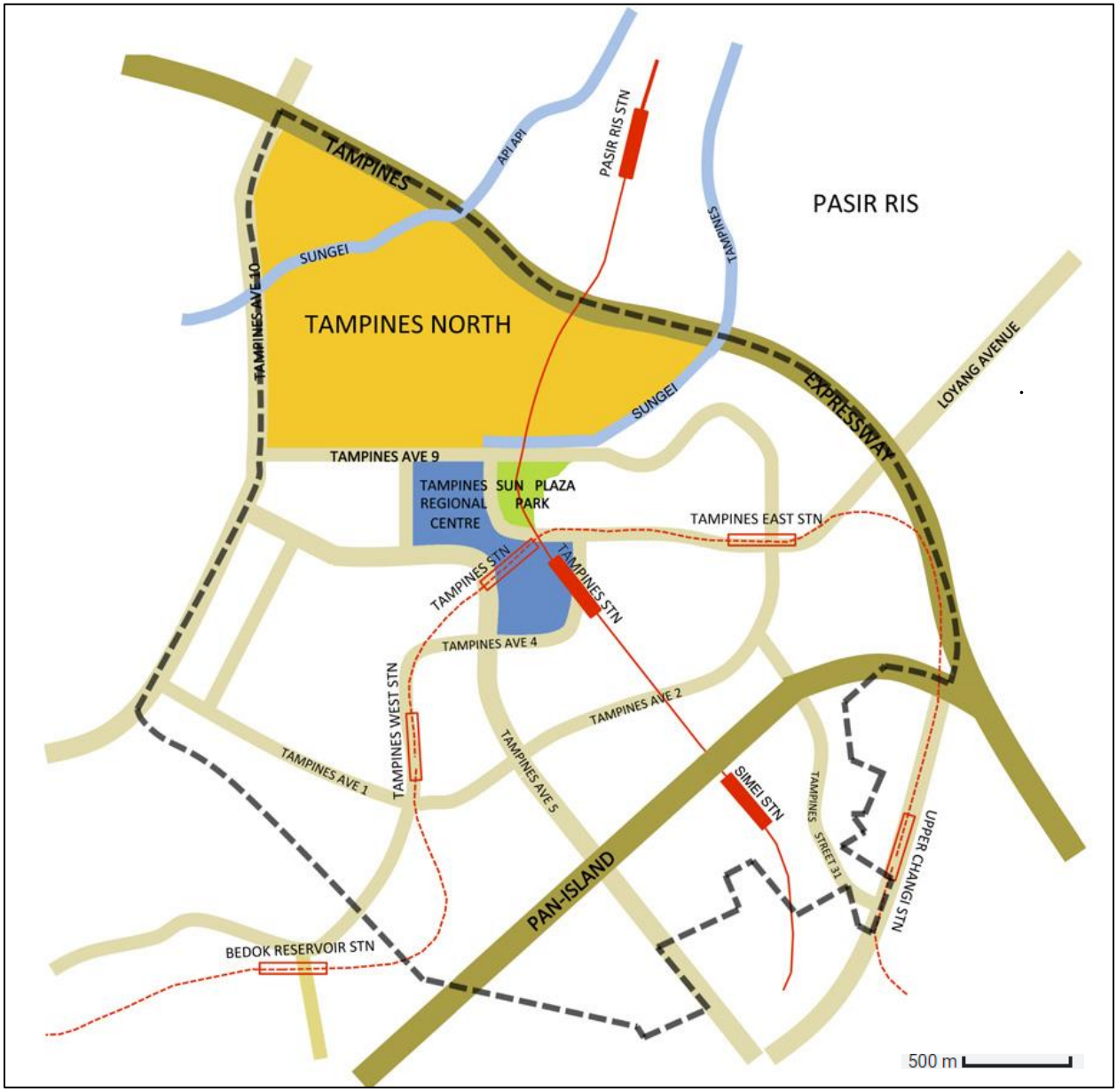
READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

This document consists of **11** printed pages, and **1** blank page.

Resource 1 for Question 1

Map of Tampines, Singapore



Resource 2 for Question 1

Group 1's survey site



Group 2's survey site



Resource 3 for Question 1

Survey results of Questionnaire

	We are a group of students conducting a survey to understand people's environmental lifestyle choices in response to climate change. Please take some time to answer these questions. **Q1 How often do you eat beef?**			
Fieldwork Group in pairs	More than 4 times a week	2-3 times a week	Once a week	Never
Group 1	5	2	3	0
Group 2	4	3	2	1
Group 3	4	3	2	1
Group 4	2	0	6	2
Group 5	2	3	5	0
Group 6	1	5	4	0
Group 7	1	0	7	1
Group 8	2	4	3	1
Group 9	1	5	3	2
Group 10	2	6	2	0
Total/100				

Q2 Have you ever chosen to use non-motorised transport options (e.g. cycling or walking)?			
	Yes, often	**Once or twice**	**Never**
Total/100	8	30	62

Q3 How often do you buy local produce?			
	Regularly	**Occasionally**	**Never**
Total/100	35	51	14

Q4 Do you use a refillable water bottle?			
	Always	**Sometimes**	**Never**
Total/100	18	71	11

Q5 How often do you use your recycling bin at home?			
	Regularly	**Occasionally**	**Never, we don't have one**
Total/100	51	39	10

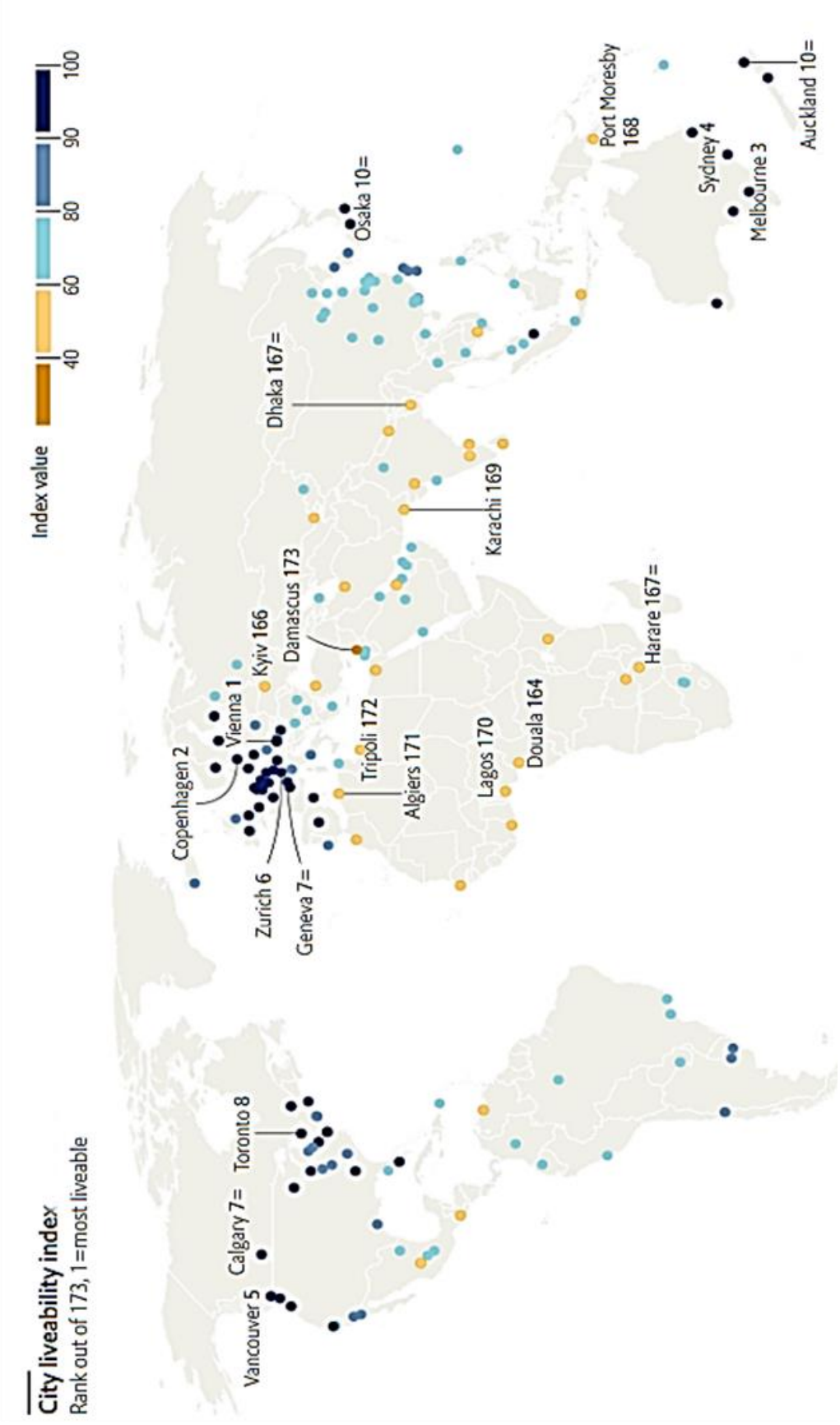
Resource 4 for Question 1

Sorting of waste in one respondent's home



Resource 5 for Question 2

City liveability index in 2023



Resource 6A for Question 2

Before Urban Reimaging of Birmingham's City Centre



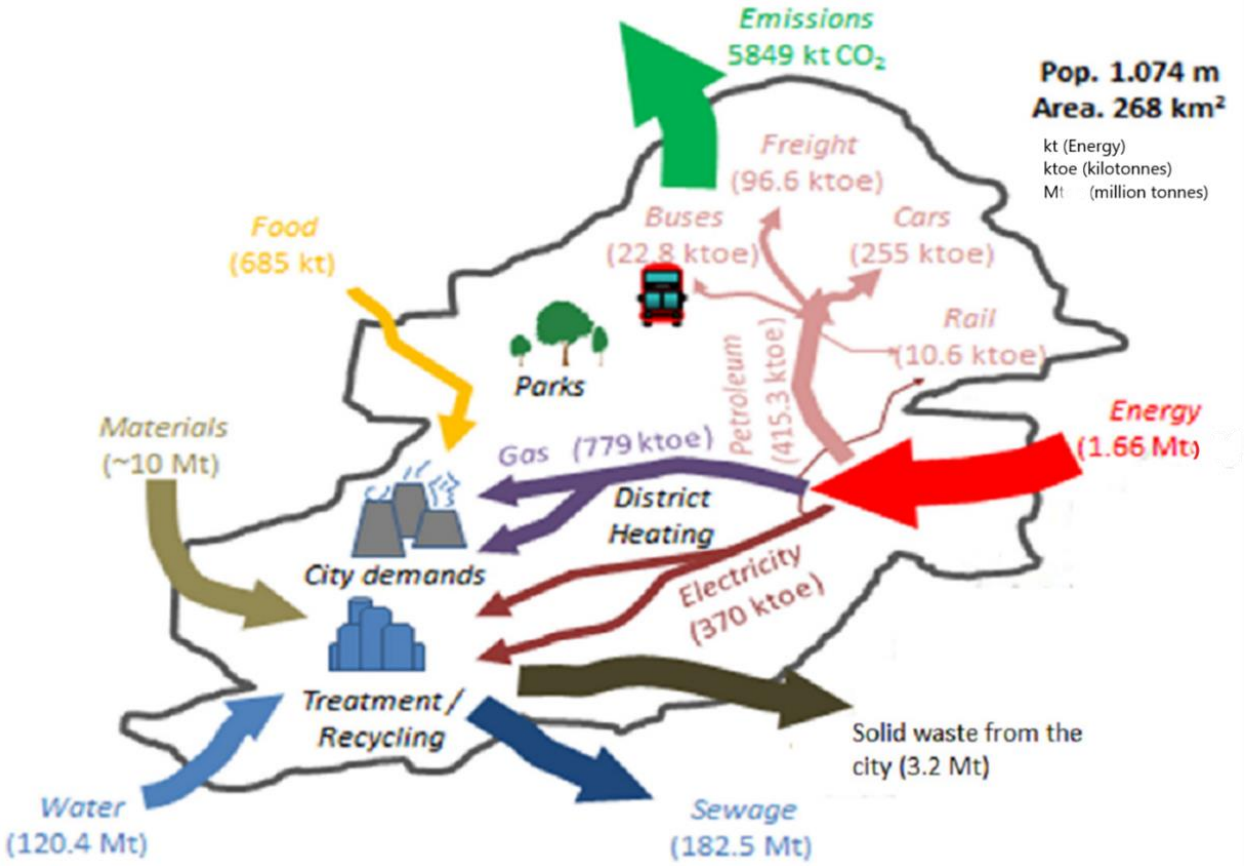
Resource 6B for Question 2

After Urban Reimaging of Birmingham's City Centre



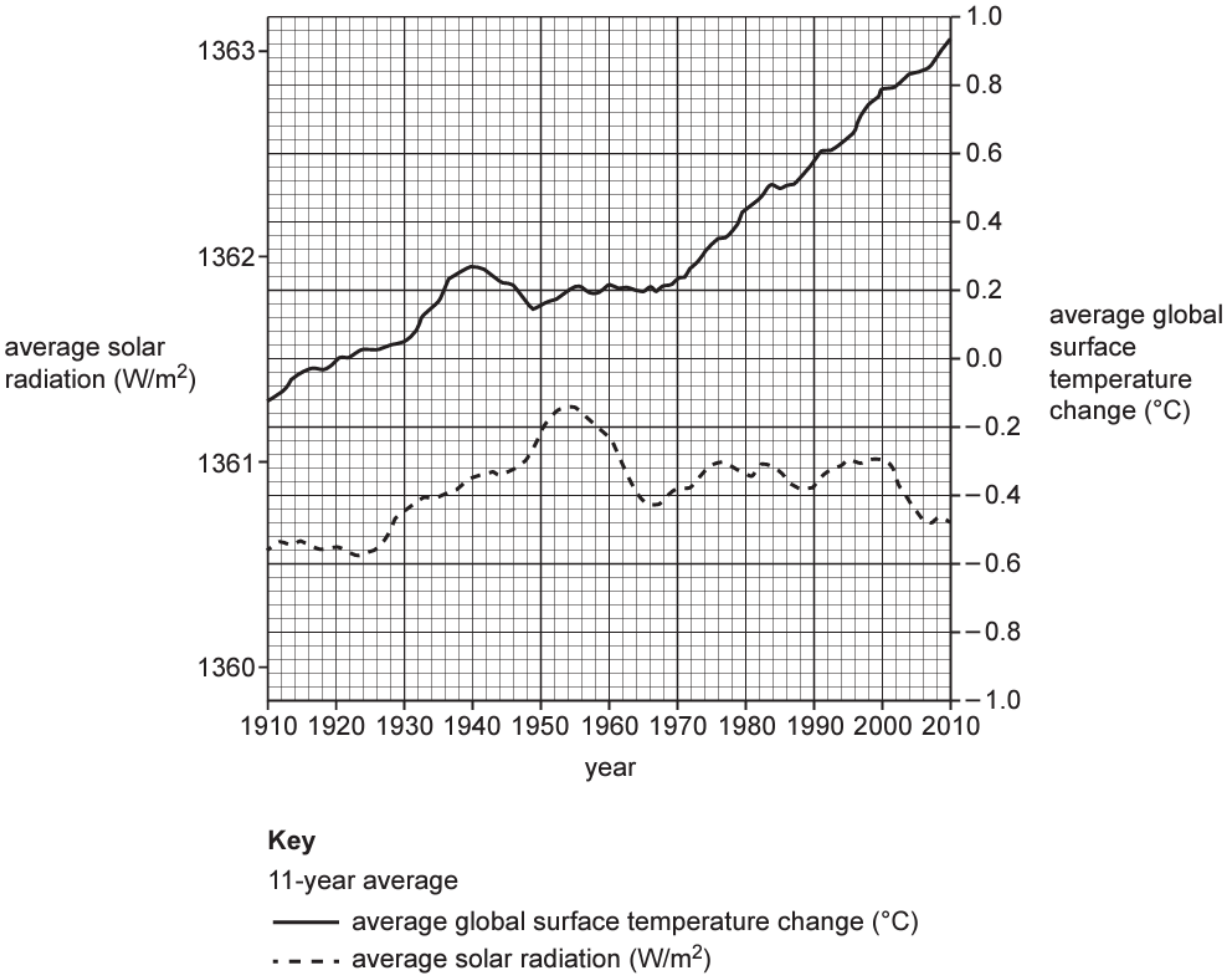
Resource 7 for Question 2

Resource flows depicting urban metabolism of Birmingham (2012-2013)



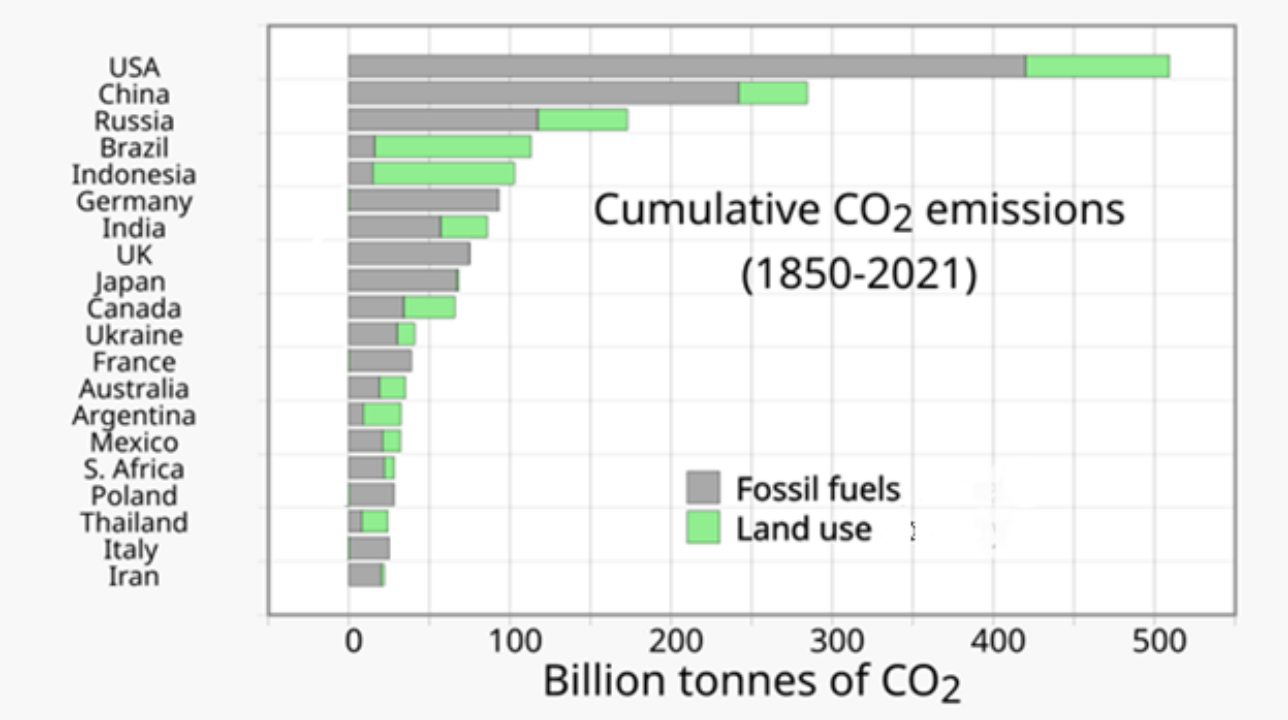
Resource 8 for Question 3

Average solar radiation and average global surface temperature change, 1910-2010



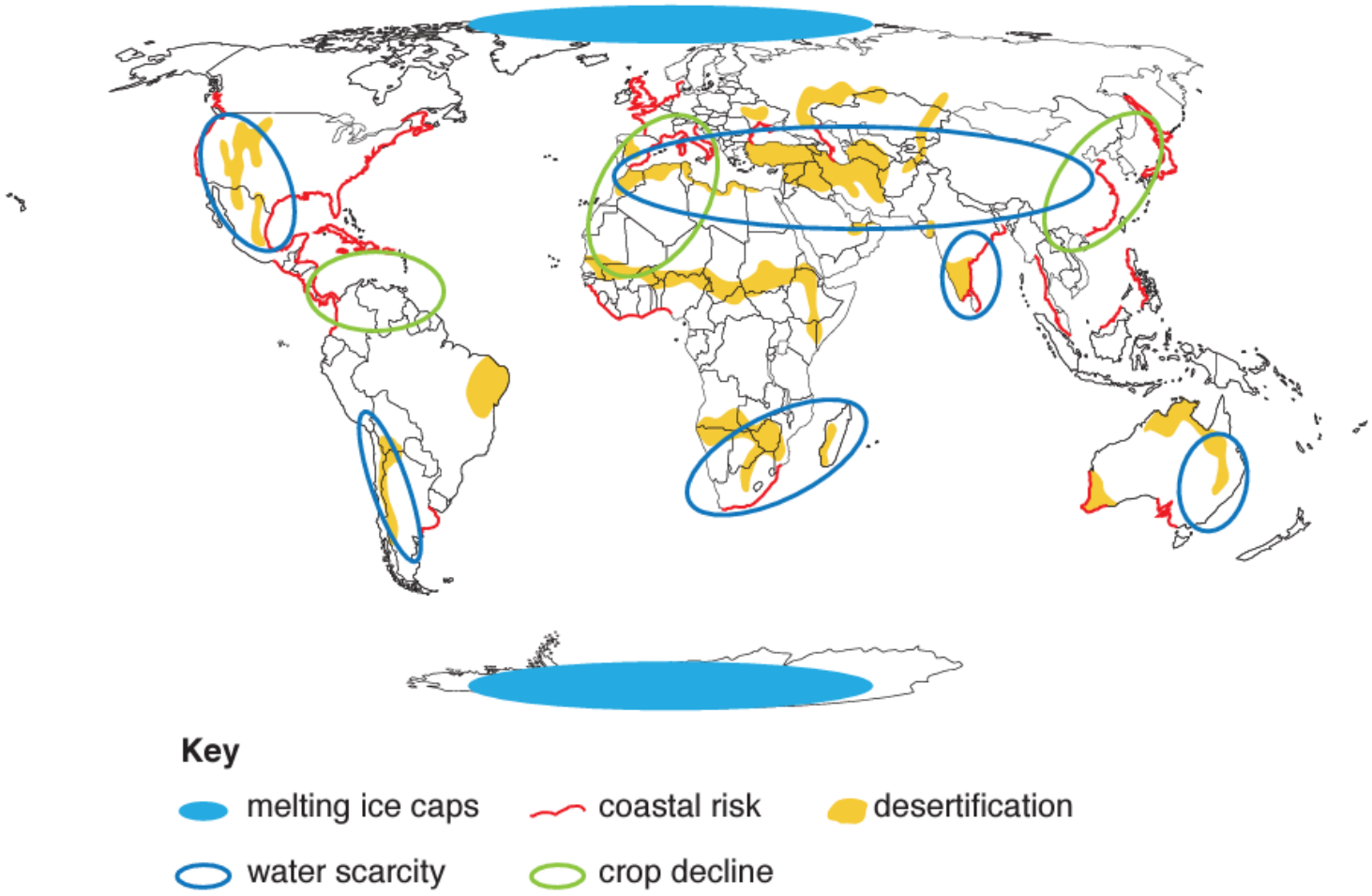
Resource 9 for Question 3

Top emitter countries and their major sources of CO_2 emissions



Resource 10 for Question 3

Some possible consequences of contemporary climate change



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